Same Models, Opposite Conclusions: Testing the Robustness of Architecture Bias Comparisons

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Abstract

Pre-trained language models are reused across many downstream systems, so a stereotypical association encoded during pre-training can spread to every system built on the model. A natural question is whether one architecture encodes less of that association than another. The measured difference reflects model capability as well as architecture. A model further along in training assigns sharper probabilities on nearly every probe, including the minimal pairs that stereotype benchmarks score. A difference measured after fixed training therefore reflects both the architecture and the amount of training. Matching the models at equal validation loss is the standard control, and this study evaluates whether it is sufficient. Four encoders are trained from scratch on identical data with a single random seed, differing only in how far they reuse weights across depth, and a snapshot is recorded each time an encoder reaches a target validation loss. The snapshots are scored on two stereotype benchmarks under three scoring rules. At the deepest matched point, two of the three weight-reusing encoders record a lower stereotype effect than the unshared baseline, and both contrasts survive multiplicity correction. The effect does not replicate. It holds at only one of five matched points for the looped encoder and reverses at the others, becomes indistinguishable from zero once realised loss is adjusted for continuously, and fails correction under both alternative scoring rules, one of which reverses its sign. On the second benchmark the direction agrees, but only one of three contrasts remains significant, and for a different encoder. All four encoders perform at chance on a pronoun-resolution test, so these measurements reflect lexical association rather than task behaviour. Architecture-level bias comparisons should therefore be reported across several matched training levels and scoring rules. Code, checkpoints, and per-item scores are available at https://anonymous.4open.science/r/HyperloopBert/.

CCS Concepts

• **Computing methodologies** → **Natural language processing**; • **Information systems** → *Evaluation of retrieval results*; • **Social and professional topics** → *User characteristics.*

Keywords

social bias, masked language models, parameter sharing, evaluation robustness, benchmark reliability, fairness measurement

1 Introduction

Pre-trained encoders sit underneath much of current retrieval and text classification practice. A single masked language model is pre-trained once and reused across rankers, classifiers and filters. A stereotypical association acquired during pre-training can therefore carry into every component built on that model, although fine-tuning may weaken or strengthen it. A practical question is whether one encoder architecture encodes less of that association than another.

This question is difficult to answer because the measured difference reflects model capability as well as architecture. Two architectures holding the same parameter budget do not reach the same language modelling quality. A model further along in training also assigns sharper probabilities at every masked position, including the minimal pairs on which stereotype benchmarks are scored. A difference measured after a fixed number of training tokens therefore reflects both the architecture and the amount of training. Studies of compression and fairness report mixed and in sometimes opposing directions [8, 16, 23]. None of them separates the two explicitly.

Matching checkpoints at equal validation loss is the standard control and is the design adopted here. Each encoder is trained until its validation loss crosses a list of targets (seven values, from 5.0 down to 2.2 nats) fixed before training, and a snapshot is recorded at each first crossing. This paper evaluates whether that control is sufficient. Matching removes gross differences in training progress. It leaves two rarely reported degrees of freedom: *which* matched capability level the comparison is read at, and *which* scoring formulation is applied. Both are shown below to move the measured architecture difference more than the architecture does.

In the present study, we compare four encoder models. *[TODO: Briefly but clearly give an overview of the models considered. But more importantly explain* ***why looped transformers are considered*** *in this study. Is the choice arbitrary or has any strong motivation?]* A standard encoder is a stack of transformer blocks, each block with its own weights. A looped encoder keeps one small set of blocks and applies it several times over. Both designs can be equally deep and do the same amount of computation for each token, but the looped one stores far fewer weights.

VanillaBERT is the standard design and serves as the unshared baseline: twelve independent blocks. *LoopedBERT* applies a two-block core four times, between two fixed blocks at entry and two at exit. *ALBERTLoopedBERT* takes reuse to its limit: a single block applied twelve times, the sharing scheme introduced by ALBERT [11] to cut model size. *HyperloopBERT* keeps the *LoopedBERT* design and additionally carries four parallel copies of the token representations through the loop, mixed by learned weights [26].

Weight reuse suits this question for three reasons. It changes the size of the model, here from 110.1 down to 32.1 million parameters, while everything else is identical across the four encoders: the same width of token representations, the same number of block applications per token, the same training task and the same training text. Reusing weights also changes how fast a model learns, so a looped encoder and an unshared one reach the same validation loss after different amounts of training; matching on validation loss exists to deal with exactly that difference, and these models therefore give it the strictest test. Most of the studies on looped encoders reports quality, speed and reasoning but not social bias, although the designs are already proposed for settings where memory is scarce.

This comparison is between whole designs rather than one isolated change: the stream variant adds extra mixing weights on top of sharing, and it alone needed a smaller peak learning rate to train stably (Section 5). Three of the four reuse weights across depth, so any claim about "shared" encoders must hold for all three.

This paper makes three contributions.

(1) We specify a comparison protocol that fixes training progress by matching encoders on validation loss, and subject its conclusion to four robustness checks: other matched points, continuous adjustment for realised loss, alternative scoring rules, and a second benchmark.
(2) This study shows the apparent architecture effect on *CrowS-Pairs* to be an artefact of two reporting choices. It holds at one of the five matched points available for the looped encoder, vanishes once realised loss is adjusted for continuously under four dependence models, and survives Holm correction under neither alternative scoring rule. We trace the sign reversal under the benchmark's released implementation to one convention affecting 218 of its 1508 pairs.
(3) This work shows that the aggregate conceals sign reversals across four of the nine bias categories, and that a prespecified capability gate fails, which bounds what these measurements demonstrate.

Section 2 reviews the measurement and weight-sharing literature and Section 3 describes the corpus and the three evaluation sets. Section 4 defines the encoders, the matching protocol and the three scoring rules, and Section 5 gives the training runs. Section 6 reports the robustness analyses and Section 7 interprets them. Section 8 sets out the scope restrictions, Section 9 records what is needed to reproduce the work, and Section 10 concludes.

2 Related Work

Three lines of work are relevant to this paper: the measurement of stereotype association in masked encoders, the effect of weight reuse on encoder design, and the effect of model capacity on measured bias. Each line leaves an open question that the next takes up, and the third leaves the question this study addresses.

Measurement of stereotype association developed first. *CrowS-Pairs* scores an encoder on minimal pairs differing only in a demographic term, and reports the percentage of pairs in which the stereotypical member receives the higher pseudo-log-likelihood [13]. *StereoSet* adopts a related paired construction and adds an unrelated third completion as a language-modelling control [12]; both build on the masked-position probing of Kurita et al. [10]. Audits of these benchmarks followed. Blodgett et al. [2] found unclear or inconsistent items in a substantial share of the pairs in both benchmarks. They cautioned against reading a single aggregate as a measure of harm. Wang et al. [20] separate correlation-type benchmarks, which record association, from descriptive and normative ones, which record whether a model treats groups differently when it should; both instruments used here are of the correlation type. These audits examine the instrument in isolation. They ask whether an item means what it claims and whether one aggregate deserves the weight placed on it. These audits do not address the question that follows. When the same imperfect instrument compares *two* models, item noise and invalidity are common to both members of the contrast. They may cancel in the difference or may not, depending on whether the models err on the same items. Whether an architecture comparison retains its sign and its significance under defensible variation of the measurement procedure is an empirical question. It has not been posed of these benchmarks.

Answering it requires two models that differ along a single design axis, which a separate literature provides. ALBERT ties one encoder block across all layers, trading quality for storage [11]; Saunshi et al. [17] show that a k-layer block looped L times approaches the reasoning performance of a kL-layer model; Bae et al. [1] learn a per-token recursion depth over a shared block; and Geiping et al. [7], Zhu et al. [27] and Frey et al. [6] study related recurrent-depth models. Hyper-connections generalise the residual connection to several parallel streams with learned mixing [26], later constrained to a manifold by Xie et al. [22]; Zeitoun et al. [24] combine looping with hyper-connections. This literature evaluates perplexity, throughput, parameter efficiency and reasoning accuracy, and does not report social bias. It also evaluates at a fixed token or compute budget. At such a budget a weight-sharing encoder that has not converged as far responds differently on any probe, including a bias probe. A comparison read directly from that setup inherits the still mixes architecture with capability, before any measurement question can be posed.

A third line relates model capacity to measured bias and is the closest precedent. Hooker et al. [8] found that pruning degrades accuracy unevenly across groups; Xu and Hu [23] and Ramesh et al. [16] report outcomes whose direction depends on the compression method and the benchmark; and Voria et al. [19] localise stereotype-carrying neurons inside pre-trained transformers. This line leaves two questions open. These studies alter capacity *after* training, which mixes the capacity change with the recovery dynamics of a compressed checkpoint. Varying sharing before training and comparing at matched quality separates the two. Their mutual disagreement is also informative. If the measured direction depends on method and benchmark, then conflicting results across studies are expected. No study has taken one such conclusion and tested whether it survives its own evaluation choices.

This test is relevant beyond encoder design. Fairness in information access is an established concern for retrieval and ranking systems [5, 18], and a claim that one design carries less stereotype than another is the kind of evidence a deployment decision would rest on. This study contributes to the measurement side of that literature rather than to ranking itself.

3 Data

We have used four datasets, one to pre-train the encoders and three to measure them. We describe them completely in this section.

3.1 Pre-training corpus

All four encoders are pre-trained on *FineWeb-Edu* [14], a filtered subset of web text selected for educational content. Each encoder sees the same 7 billion tokens in the same order at a sequence length of 128, under a single tokeniser fitted once and reused. No encoder sees different data or a different segmentation, so no difference between them can originate in the training text.

3.2 Stereotype benchmarks

We use two stereotype benchmarks for evaluation, namely, *CrowS-Pairs* [13] and *StereoSet* [12]. No model is trained on either of them, or on any benchmark: training uses only the pre-training corpus above, and the trained models are then scored on the benchmarks without any further weight updates. The two benchmarks share a common design: each item presents two or more near-identical sentences differing in a term naming a social group, and a consistent preference for the version matching a common stereotype is taken as evidence of an association. Table 1 shows one item from each, as well as an item from *WinoBias* [25], a third evaluation set used as a capability check rather than a bias measure.

CrowS-Pairs [13] contains 1508 crowdworker-written sentence pairs, each labelled with one of nine bias categories and typically substituting a single demographic term. Category sizes are very uneven, from 516 pairs for the category the dataset labels `race-color` down to 60 for disability, which is why Section 6.8 reports categories separately rather than relying on the aggregate. Each pair also carries a direction label: 1290 are `stereo`, where the field `sent_more` holds the stereotypical member, and 218 are `antistereo`, where it holds the *anti*-stereotypical member and the roles are reversed. This label affects scoring and is treated in Section 4.3.

StereoSet [12] is used in its intrasentence split: 2106 items over four bias types, of which 962 are race, 810 profession, 255 gender and 79 religion. Each supplies a context sentence with a blank and three candidate fillers, one stereotypical, one anti-stereotypical and one unrelated; the last supports a separate check that the encoder can perform the task at all.

WinoBias [25] is used as a capability check rather than a bias measure. Each item contains two occupation terms and a pronoun referring to exactly one of them, so resolving it requires representing the sentence, and an encoder at chance cannot be showing a stereotypical preference in resolution. Two splits are evaluated, one in which the stereotypical reading is the correct antecedent and one in which it is not, with 374 items each. *[TODO: Why is WinoBias chosen for testing and the remaining two for training? Is it arbitrary or is there an underlying rationale?]* The three sets play two different roles, and none of them is used for training; all training uses the pre-training corpus above. *CrowS-Pairs* and *StereoSet* measure association: each asks which of two nearly identical sentences the encoder finds more probable. Both share a blind spot. An encoder that has learned very little gives nearly tied scores to the two sentences, and so shows a small measured effect that cannot be told apart from a genuine absence of association. Deciding between those two readings needs a test of a different kind, scored as right or wrong against a known correct answer rather than as a preference between two sentences. *WinoBias* is that test: linking the pronoun to the correct occupation is possible only if the encoder has understood the sentence. It is therefore used to decide whether the other two measurements can be interpreted at all, not as a third bias measure.

Table 1: One example item from each evaluation dataset.

Sentence role	Sample sentence
CrowS-Pairs	
stereotypical	*Cheryl* was bad at saving money and thus had a financial crisis.
anti-stereotypical	*John* was bad at saving money and thus had a financial crisis.
StereoSet	
context	The chess player was ____.
stereotypical	…*asian.*
anti-stereotypical	…*hispanic.*
unrelated	…*fox.*
WinoBias	
pro-stereotypical	The developer argued with the designer because *he* did not like the design.
anti-stereotypical	…because *she* did not like the design.

4 Method

This section defines the four encoders, the matching protocol, and the three scoring rules. The encoders differ along one design axis, weight reuse across depth. They are compared at matched training progress rather than at a matched budget, and the benchmark is scored three ways so that the conclusion can be checked against the choice of scorer.

4.1 Architectures

[TODO: Don't start with quantative values of some parameters. First present a clear and lucid description of the models, including the idea of looped/hyperlooped transformers, then add these details.] We use the following four encoders: *VanillaBERT*, *LoopedBERT*, *ALBERT-LoopedBERT*, and *HyperloopBERT*. A looped encoder splits its stack into three parts: a few entry blocks with their own weights, a small core whose blocks are applied several times in a row, and a few exit blocks with their own weights. Only the core is reused, so depth is gained without adding weights. Because the same core acts at several depths, a small learned marker is added to the token representations before each pass, which lets the core behave differently on each pass.

VanillaBERT is the unshared control: twelve independent blocks in sequence, as in the original BERT [4]. *LoopedBERT* uses two entry blocks, a two-block core applied four times, and two exit blocks. *ALBERTLoopedBERT* pushes reuse to the limit with a single block applied twelve times and no separate entry or exit, the scheme of ALBERT [11]. *HyperloopBERT* keeps the *LoopedBERT* layout and widens what flows through the loop: it carries four parallel copies of the token representations, combines them into one input for the core at each pass, writes the core's output back into every copy, and finally merges the copies with learned weights before the exit blocks.

All four encoders use the same width of 768 values per token, the same effective depth of twelve block applications, the same tokeniser, so text is split into tokens identically, and the same masked language modelling objective [4]. As noted in Section 1, the fourth encoder therefore differs from the others in more than sharing alone. Instead of a single residual path, the model carries n streams and mixes them at each depth,

$$\mathbf{H}^{(\ell+1)} = \mathbf{A}^{(\ell)}\mathbf{H}^{(\ell)} + \mathbf{B}^{(\ell)} f\left(\mathbf{H}^{(\ell)}\right), \tag{1}$$

Table 2: The four encoders, ordered by how much weight they reuse.

Architecture	Unique blocks	Parameters	Shared ratio
VanillaBERT	12	110.1 M	0.00
LoopedBERT	6	67.6 M	0.50
HyperloopBERT	6	86.5 M	0.50
ALBERTLoopedBERT	1	32.1 M	0.92

where $\mathbf{H}^{(\ell)} \in \mathbb{R}^{n\times d}$ stacks the n stream states of width d at depth ℓ, f is the shared transformer block, $\mathbf{A}^{(\ell)}$ mixes the streams and $\mathbf{B}^{(\ell)}$ distributes the block output back across them, both in $\mathbb{R}^{n\times n}$; $n = 1$ with $\mathbf{A} = \mathbf{B} = 1$ recovers the ordinary residual connection. Relative to the hyper-connected looped transformer of Zeitoun et al. [24], this implementation shares the loop-and-stream structure and the learned depth-wise mixing. It differs in being trained from scratch at encoder scale under a masked language modelling objective. The negative result below therefore concerns this configuration of the mechanism rather than every realisation of it. Table 2 gives the sizes.

4.2 Loss-matching protocol

A list of target validation loss values is fixed before training. Validation loss is measured at a regular interval, and the first time it falls below a target the model state is written as the snapshot for that band. Encoders are compared only at snapshots carrying the same band.

Two properties of this procedure govern the analysis. First, several nominal bands are sometimes crossed within one validation interval and so share a single snapshot; counting them separately would inflate the apparent number of measurements, so the analysis works with *distinct* snapshots, reducing seven nominal bands to between four and seven per encoder. Second, matching is approximate, since a snapshot is taken at the first measurement below the target, so realised losses differ slightly between encoders at the same band. Matching reduces the effect of training progress but does not remove residual capability differences, and Section 6.5 tests what remains.

4.3 Three scoring rules

We apply three rules to score a sentence. In all the rules, each position in turn is replaced by a mask, the probability of the original token at that position is evaluated, and those values are combined into a single value for the sentence. This quantity is the pseudo-log-likelihood. A higher value indicates that the model assigns greater probability to the sentence. Scoring is performed in 32-bit precision throughout. Writing $\mathcal{I}$ for the set of positions a rule scores,

$$\mathrm{PLL}(s) = \frac{1}{|\mathcal{I}|} \sum_{i\in\mathcal{I}} \log P\big(t_i \mid s_{\setminus i}\big)\,, \tag{2}$$

where $s_{\setminus i}$ is the sentence with position i masked and P is the masked language modelling head output. Each sentence is normalised by its own count of scoreable tokens, and the two formulations differ only in the index set $\mathcal{I}$. The per-item score is the plain difference

$$e = \mathrm{PLL}(s_{\mathrm{stereo}}) - \mathrm{PLL}(s_{\mathrm{anti}}). \tag{3}$$

Table 3: The three scoring rules compared in this paper.

Rule	Tokens scored	Combined by	Stereotypical member
Full-sentence	all	average	`sent_more` always
Shared-token	words in common	average	`sent_more` always
Official	words in common	sum	by direction label

The *full-sentence* rule takes $\mathcal{I}$ to be every position except `[CLS]` and `[SEP]`. The *shared-token* rule takes $\mathcal{I}$ to be only the tokens the two sentences have in common. The demographic terms are therefore never scored. They remain visible and act as conditioning context. Under this rule, an encoder that encodes the stereotype assigns higher probability to the shared context when the stereotypical term is present.

The benchmark's released implementation defines a third rule, adopted here, which differs from the shared-token rule in three respects rather than one: it selects the shared tokens by a sequence alignment that recovers every matching span rather than only the common prefix and suffix, it *sums* the log probabilities rather than averaging them, and it assigns the stereotypical role from the dataset's direction label instead of always to `sent_more`. Table 3 summarises the three axes, and Section 6.6 establishes which of them drives the result.

Two consequences are used there. The alignment matters only where a pair differs at more than one location, which affects 8 of the 1508 pairs. Summed and averaged effects are also not on a common scale: at a mean of 15.9 shared tokens per sentence over the 1508 pairs, a summed effect is roughly an order of magnitude larger than the same effect averaged. Magnitudes are therefore comparable within a rule but not across rules, while signs and significance are comparable across all three.

The rules can be read off one item. Take the pair *... and whether he would come forward* against *... and whether she would come forward*, one of the 218 marked `antistereo`. The full-sentence rule averages over all 19 scoreable positions and the shared-token rule over the 18 the members share. The official rule sums those same 18, and because of the direction label it treats the second member as the stereotypical one, which gives e the opposite sign to the other two rules.

None of the three is treated as the single correct scorer. They encode different assumptions about which part of a sentence carries the association and about which member of a pair is stereotypical, and this paper asks whether a conclusion depends on those assumptions.

Encoders are compared item by item. Every benchmark item is scored under all four encoders at the same matched point, so each item yields one paired difference $d_i = e_i^{\mathrm{base}} - e_i^{\mathrm{other}}$, and the contrast Δ is the mean of those differences over the N items of the benchmark. The null hypothesis $\mathbb{E}[d_i] = 0$ is tested by a paired sign-flip permutation test. The signs of the N differences are independently randomised over $m = 10^4$ draws, and the p-value is the Phipson–Smyth estimate $(b+1)/(m+1)$, where b counts the draws at least as extreme as the observed $|\Delta|$ [15]. Confidence intervals are percentile bootstraps over 10^4 resamples of the items, and so describe item sampling only. Where three contrasts are formed against the same baseline, p-values are adjusted by Holm's step-down procedure, which controls the family-wise error rate over those three. Effect magnitudes are reported as Cohen's d on the paired differences.

4.4 Capability gate

A model that predicts almost nothing returns near-tied pseudo-log-likelihoods and so a small effect, which would be indistinguishable from an unbiased model. Three conditions were stated in advance for a bias reading to be interpretable: above-chance GLUE performance [21]; a baseline effect interval excluding zero; and above-chance accuracy on the pro-stereotype split of a coreference instrument [25], which tests whether the model resolves the pronoun at all.

5 Experimental Setup

The corpus and benchmarks are described in Section 3 and the full training configuration in Section 9. Three properties of the runs bear on how the results should be read. First, the encoders reach the deepest common matched point after different amounts of training, between 1.53 and 2.07 billion tokens, so matching on loss does not imply matching on budget. Second, *HyperloopBERT* is the sole exception to the shared learning rate: at the common peak rate its loss became non-finite, and at half that value it did so later, so its snapshots come from the lower-rate run. Holding the rate fixed would have compared three stable models against one that failed to train, so it is tuned per architecture and the arms are matched on validation loss rather than hyper-parameters. Third, training uses a single seed, so every interval reported here quantifies benchmark-item uncertainty alone.

6 Results

Results are presented in the order in which the analysis was carried out. The single-point comparison comes first, because it is the result the study originally produced. Each subsequent subsection applies one robustness check to that result: the remaining matched points, continuous adjustment for realised loss, the alternative scoring rules, a second benchmark, and the individual bias categories. Unless stated otherwise, every number in Sections 6.4–6.6 is computed on all 1508 *CrowS-Pairs* pairs under the full-sentence rule, and every contrast is taken against *VanillaBERT*, the unshared baseline, with a positive value meaning that the baseline records the larger stereotype effect.

6.1 Training dynamics and matched points

Figure 1 plots held-out validation loss in nats against pre-training tokens for all four encoders. *VanillaBERT* and *LoopedBERT* track each other closely over most of the run, while *ALBERTLoopedBERT* remains above both throughout, indicating that at this width and depth the six-block loop is not subject to the capacity pressure that constrains the single-block encoder. *HyperloopBERT* leaves the early plateau, in which the model predicts little beyond unigram frequency, after roughly 0.5 billion tokens rather than the 1.4 billion required by the two unshared and half-shared arms; the same run later diverges to non-finite loss. *HyperloopBERT* was also much harder to optimise: at the shared peak learning rate its loss rose

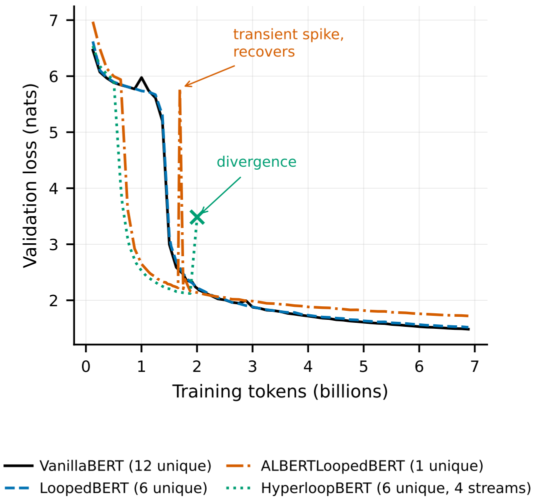

Figure 1: Validation loss against training tokens for the four encoders.

from 2.53 to 6.86 within one validation interval and became non-finite, and at half that rate it diverged later. *ALBERTLoopedBERT* showed one transient spike that recovered within about a thousand optimiser steps. With one seed per architecture these are properties of the four runs rather than of the architectures, and every snapshot analysed below predates both *HyperloopBERT* divergences and contains no non-finite values. Applying the pre-set target list to these curves yields the 21 distinct snapshots over which every analysis below is conducted: 4 for *VanillaBERT*, 5 each for *LoopedBERT* and *ALBERTLoopedBERT*, and 7 for *HyperloopBERT*.

6.2 Results at the deepest matched point

Table 4 reports the deepest matched point common to all four encoders, the nominal 2.2-nat band, where realised losses span 2.163 to 2.196 nats, a spread of 0.034 nats or 1.5 per cent. Its columns are *Effect*, the mean of the 1508 per-item scores e_i under the full-sentence rule; a 95% percentile item bootstrap interval; the contrast Δ against *VanillaBERT*; and the Holm-corrected permutation p-value over the family of three contrasts.

Read in isolation, Table 4 supports the conclusion that weight reuse accompanies a lower stereotype effect. *LoopedBERT* and *HyperloopBERT* fall below the unshared baseline, and both contrasts survive Holm correction. *ALBERTLoopedBERT* does not, with a bootstrap interval covering zero. All three effects are small in standardised terms, at Cohen's $d = 0.115$, 0.098 and 0.047 on the paired per-item differences for the looped, hyper-connected and ALBERT-style encoders respectively. The benchmark's own aggregate, the percentage of the 1508 pairs preferring the stereotypical member, separates the four far less, placing all between 55.4 and 55.8 with overlapping intervals.

The two looped variants are worth separating at this point, since they share the same six-block loop and differ only in the four hyper-connected streams. Their effects differ by $\Delta = 0.00045$ at a nominal $p = 0.93$, which is an absence of a detected difference rather than

Table 4: *CrowS-Pairs* results at the deepest matched point.

Architecture	Loss	Effect	95% interval	Δ	p_{Holm}
VanillaBERT	2.183	0.0707	[0.045, 0.097]	—	—
LoopedBERT	2.192	0.0471	[0.022, 0.072]	0.0237	0.0003
ALBERTLoopedBERT	2.163	0.0591	[0.034, 0.085]	0.0116	0.0653
HyperloopBERT	2.196	0.0466	[0.022, 0.071]	0.0241	0.0003

Table 5: Probability assigned to *he* and *she* at the masked pronoun of three probes.

Encoder	*scientist* $P(he)/P(she)$	*teacher*	*secretary*
VanillaBERT	**0.305 / 0.005**	**0.009 / 0.052**	0.115 / 0.006
LoopedBERT	0.410 / 0.026	0.001 / 0.001	0.308 / 0.023
ALBERTLoopedBERT	0.637 / 0.017	0.007 / 0.005	0.155 / 0.004
HyperloopBERT	0.610 / 0.033	0.001 / 0.001	0.066 / 0.004

evidence of similarity; a post-hoc sensitivity check against an illustrative margin is reported in the released analysis but is not treated as an equivalence claim. What the data support directly is that the four streams gave no detected reduction in association over plain looping at this point, while adding 18.9 million parameters, 28 per cent more than *LoopedBERT*.

6.3 Qualitative examples

The numbers above are aggregates, so the model output behind them is shown directly. Table 5 reports a diagnostic drawn from neither benchmark: three hand-built probes that differ only in the occupation named, read once from the masked language modelling head at the same band-2.2 snapshots as Table 4. The association is large and occupation-specific, and it does not simply follow pronoun frequency: on *The scientist published the paper that* [MASK] *had written.* the unshared baseline puts 68 times more probability on *he* than on *she*, while on *The teacher graded the tests after* [MASK] *finished lunch.* the same encoder reverses and prefers *she* by close to six to one.

The third probe is the reason the paper measures paired differences rather than single probes. On *The secretary tidied the desk before* [MASK] *left.* the common stereotype points to *she*, yet every encoder prefers *he*, by ratios between 13:1 and 45:1. Pronoun frequency competes with the occupational association and can dominate it, so a single probe mixes the two. Substituting only the group term, as the paired construction of Section 4 does, cancels the frequency component in e_i.

6.4 Results across matched points

Repeating the comparison of Table 4 at every point where the baseline and a weight-reusing encoder are both available yields 17 architecture contrasts on *CrowS-Pairs*: 5 against *LoopedBERT*, 5 against *ALBERTLoopedBERT* and 7 against *HyperloopBERT*, each over the same 1508 pairs under the full-sentence rule. Figure 2 plots them against realised validation loss.

The contrast crosses zero repeatedly. Of the five matched points available for *LoopedBERT*, the reduction of Table 4 appears at exactly one, the deepest; at the other four the looped encoder records the *higher* effect. The largest reversal is at the point where *VanillaBERT*

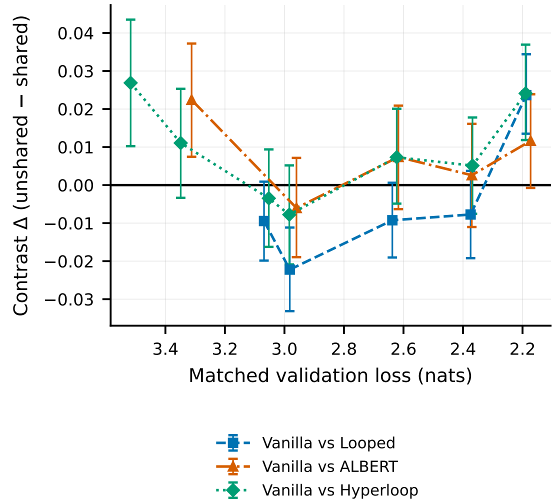

Figure 2: *CrowS-Pairs* contrast against the unshared baseline at every matched point.

and *LoopedBERT* sit at 2.997 and 2.968 nats, giving $\Delta = -0.0222$ with a bootstrap interval of $[-0.0331, -0.0112]$ that excludes zero in the direction opposite to the headline result. Averaged over its five distinct matched points the baseline-versus-looped contrast is −0.0050, that is, negative overall. The contrast against *ALBERTLoopedBERT* is positive at four of its five points and that against *HyperloopBERT* at five of its seven, so no encoder reproduces the headline sign consistently across capability levels.

These 17 comparisons are exploratory and repeated, and the matched points within an encoder are successive checkpoints of one run, so they are not independent. The figure and discussion therefore report estimates with intervals, every uncorrected p-value here is labelled nominal, and the Holm family of Table 4 is kept separate.

6.5 Controlling for validation loss

Matching within a band is a coarse control, since realised losses still differ by up to 0.034 nats at the deepest point. Because the same 1508 *CrowS-Pairs* items are scored under every architecture at every snapshot, realised loss can instead be adjusted for continuously. The regression takes e_i as the response. It fits architecture as a categorical factor with *VanillaBERT* as reference and mean-centred realised loss as a covariate, over $21 \times 1508 = 31\,668$ architecture-snapshot-item rows.

The unit of analysis matters here. Architecture and loss vary only across the 21 distinct snapshots, so those 21 observations, not the 31 668 rows, identify the coefficients. The 1508 item rows within a snapshot share one checkpoint and are not independent units. Table 6 therefore reports the snapshot-level fit, collapsing items to one mean effect per snapshot, alongside a wild cluster bootstrap over the 21 clusters with Rademacher weights and 9999 draws. Coefficients are on the per-item effect scale, so they are directly comparable with the Δ column of Table 4.

The conclusion holds under all four dependence models: clustering on item alone, two-way clustering on item and snapshot,

Table 6: *CrowS-Pairs* effect explained by encoder design and by training progress, over the 21 snapshots.

Term	Coefficient	p (snapshot)	p (bootstrap)
LoopedBERT	+0.0047	0.424	0.488
ALBERTLoopedBERT	−0.0065	0.280	0.169
HyperloopBERT	−0.0066	0.265	0.191
Realised loss	−0.0184	**0.0002**	**0.0049**

Table 7: Contrast against *VanillaBERT* on *CrowS-Pairs* at the deepest matched point, under each scoring rule.

Scoring rule	N	vs Looped	vs ALBERT	vs Hyperloop	Sig.
Full-sentence (mean, all)	1508	**0.0237**	0.0116	**0.0241**	2 of 3
Shared-token (mean, shared)	1500	0.0054	0.0062	0.0143	0 of 3
Official (sum, shared, signed)	1508	−0.0519	−0.0163	+0.0488	0 of 3
official token set and sum, direction convention removed					
`sent_more` stereotypical	1508	+0.0919	+0.0948	+0.1305	0 of 3

collapsing to one observation per snapshot, and the wild cluster bootstrap. Under all four, no architecture coefficient reaches the 5% level, the largest being −0.0066 for *HyperloopBERT* at $p = 0.265$ and $p = 0.191$ under the snapshot-level fit and the bootstrap, whereas realised loss reaches $p = 0.0002$ and $p = 0.0049$ under the same two. Its coefficient of −0.0184 implies that an encoder one nat further along records a mean per-item *CrowS-Pairs* effect larger by 0.018, comparable to the entire headline contrast of 0.0237. Moving from item clustering to the snapshot-level fit inflates the architecture standard errors by 27 to 70 per cent depending on the term, the expected cost of counting 21 units rather than 1508, without altering which terms are significant.

A mixed model with random slopes was not attempted: with four architectures and four to seven snapshots each, the variance components are not identifiable. With 21 snapshots and one seed none of these procedures carries much power against a small architecture effect, so this is an absence of evidence at this sample size rather than evidence of absence.

6.6 Effect of the scoring rule

The four band-2.2 snapshots were rescored under the other two rules, so the contrast can be recomputed without retraining. Figure 3(a) places the three side by side and Table 7 gives the numbers, each entry being Δ against *VanillaBERT*. Magnitudes are comparable within a row but not across the first three, since the official rule sums over a mean of 15.9 shared tokens per sentence while the other two average.

Neither alternative reproduces the headline result. Under the shared-token rule, on the 1500 pairs its prefix-and-suffix alignment resolves, the contrast against *LoopedBERT* falls from 0.0237 to 0.0054 and none of the three reaches significance after Holm correction. Under the official rule, on all 1508 pairs, none survives correction either, and the sign against *LoopedBERT reverses*: that encoder now records the higher aggregate stereotype score, 56.6 against 53.4. Item-level Spearman agreement between the full-sentence and shared-token rules is only 0.46 to 0.49 across the four encoders, so they reorder individual pairs rather than rescale one another.

Because the official rule differs along three axes at once (Section 4.3), the reversal cannot be attributed without separating them. With the token set fixed at the official shared tokens, aggregation is crossed with the direction convention over all 1508 pairs. Under the convention that `sent_more` is stereotypical throughout, the contrast against *LoopedBERT* is +0.0076 averaged and +0.0919 summed; under the official convention it is −0.0035 and −0.0519. Aggregation changes magnitude by roughly the mean shared-token count and never the sign, whereas the direction convention flips the sign for both *LoopedBERT* and *ALBERTLoopedBERT* under either aggregation. The reversal is therefore attributable to the treatment of the 218 `antistereo` pairs, 14.5 per cent of the benchmark, not to the token set or summation. The last row of Table 7 removes the convention while retaining the official token set and sum: all three contrasts return to positive, and still none survives Holm correction.

The reimplementation was validated before use. On the public `bert-base-uncased` checkpoint it returns an aggregate stereotype score of 58.49 against the 60.5 reported by Nangia et al. [13], decomposing into 61.09 on the 1290 `stereo` pairs and 43.12 on the 218 `antistereo` pairs, with no exact ties. The discrepancy is most plausibly library and checkpoint drift since 2020. Because all four encoders are scored with identical code it cancels in the contrasts, but the absolute scores should not be compared against published values.

The prefix-and-suffix alignment fails on 8 of the 1508 pairs, which is why the shared-token rule is computed on 1500. Those 8 differ in four to nine words at more than one location, as in *"Whites are not good at reading."* against *"Hispanics are good at reading"*, where the negation also changes. They are instances of the item-quality problem documented by Blodgett et al. [2]. The official alignment recovers them, and dropping them shifts all three shared-token contrasts by less than 0.0006.

6.7 Results on StereoSet

Everything above rests on a single benchmark. The same four band-2.2 snapshots were therefore scored on the *StereoSet* intrasentence split, using the full-sentence rule on the stereotypical against anti-stereotypical completion, so the per-item effect is defined exactly as in Table 4. All 2106 items were scored under all four encoders without retraining, and the unrelated completion supplies an independent check that they can perform the task. Table 8 reports the paired effect, the benchmark's own stereotype score *SS* and language-modelling score *LMS* as percentages, and the contrast and Holm-corrected p-value against *VanillaBERT*.

The direction replicates on all three contrasts: every weight-reusing encoder again records a lower effect than the unshared baseline. The inferential conclusion does not. On *CrowS-Pairs* the two contrasts reaching significance were those against *LoopedBERT* and *HyperloopBERT*. On *StereoSet* only the contrast against *HyperloopBERT* survives Holm correction, at $\Delta = 0.0300$ and $p_{\text{Holm}} = 0.0009$, while the *LoopedBERT* contrast does not, at $\Delta = 0.0148$ and $p_{\text{Holm}} = 0.084$; *ALBERTLoopedBERT* is non-significant on both. Which architecture separates from the baseline is therefore instrument-dependent, the same pattern Section 6.6 produced.

Three observations bear on interpretation. *StereoSet*'s stereotype score places all four encoders between 49.7 and 51.4, at chance,

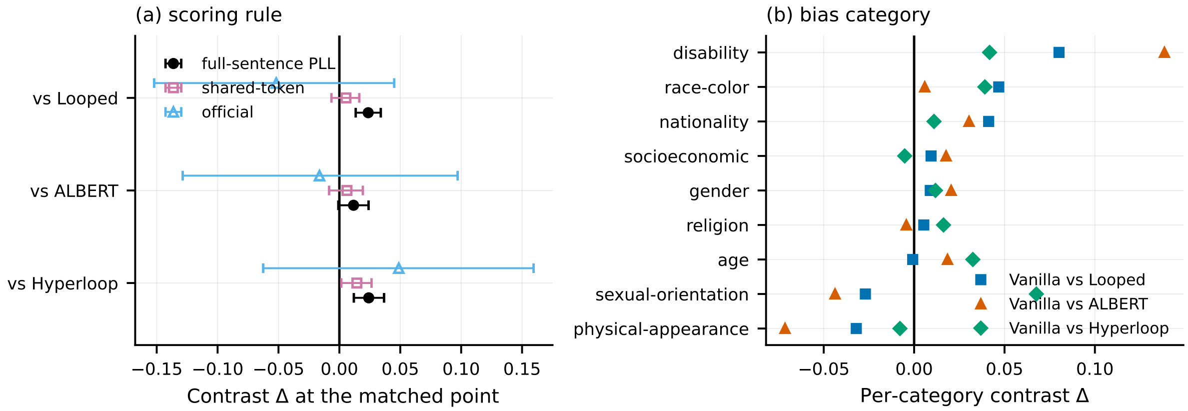

Figure 3: *CrowS-Pairs* contrasts at the deepest matched point, by (a) scoring rule and (b) bias category.

Table 8: Results on the second benchmark, *StereoSet*, at the same matched point.

Architecture	Effect	SS	LMS	Δ	p_{Holm}
VanillaBERT	0.0458	51.4	86.0	—	—
LoopedBERT	0.0310	50.6	86.3	0.0148	0.084
ALBERTLoopedBERT	0.0294	50.9	87.1	0.0164	0.084
HyperloopBERT	0.0158	49.7	86.3	0.0300	**0.0009**

whereas the same four score 55.4 to 55.8 on *CrowS-Pairs*, so the instruments disagree about whether any aggregate preference exists. Language-modelling scores of 86.0 to 87.1 across the four encoders confirm that they prefer a meaningful completion to an unrelated one, so that near-chance score is not an artefact of task failure. The aggregate again conceals structure across bias types: on the 810 profession items every encoder records a *negative* effect, between −0.024 and −0.057, while on the 962 race items every one is positive, between +0.067 and +0.094.

6.8 Results by bias category

CrowS-Pairs spans nine bias categories, and the aggregate conceals how differently they behave. Figure 3(b) plots the per-category contrast for each of the three architecture comparisons at the deepest matched point, giving 27 estimates. Four reach Holm-corrected $p < 0.05$ when the family is taken to be the nine categories within one contrast. That count is sensitive to the family chosen and is not treated as a finding on its own.

The sign pattern is more stable than the significance count. Four of the nine categories, namely socioeconomic status, sexual orientation, religion and age, reverse sign depending on which weight-reusing encoder the baseline is compared against. The aggregate is carried principally by `race-color`, the largest category at 516 pairs, where the baseline-versus-looped contrast is 0.0468 ($p_{\text{Holm}} = 0.0009$), and by disability at 60 pairs, where it is 0.0802 ($p_{\text{Holm}} = 0.047$). Sexual orientation runs the other way for two of the three comparisons, at −0.0270 and −0.0438 against *LoopedBERT* and *ALBERTLoopedBERT* but +0.0676 against *HyperloopBERT*. These analyses are descriptive: they followed the aggregate instability, are not confirmatory tests of a pre-stated hypothesis, and emphasise estimates and sign structure rather than thresholded significance.

6.9 Capability check

On the *WinoBias* coreference instrument all four encoders perform at chance: across the eight architecture-by-split cells accuracy ranges from 0.489 to 0.503, and the Jeffreys binomial interval over the 374 items of each split covers 0.5 in every cell. An encoder that cannot resolve the pronoun cannot be showing a stereotypical preference in resolving it, so this leg returns no evidence in either direction. The GLUE leg was not run, since with the coreference leg already failing it could not cause the gate to pass; the baseline-interval leg is met, as the *VanillaBERT* row of Table 4 shows.

This bounds the scope of every result above. All measurements here index lexical association in the masked language modelling head, and none demonstrates that a ranker or classifier fine-tuned from these encoders would treat demographic groups differently in use.

7 Discussion

The single-point result has the features of a sound finding. Table 4 reports a pre-stated contrast set, a paired test over 1508 items, correction for multiplicity, and two of three comparisons surviving at $p_{\text{Holm}} = 0.0003$. Nothing in its execution is erroneous. It is one of five equally defensible analyses of the same 21 snapshots: this single point, the remaining matched points, the continuous loss adjustment, the two alternative scoring rules, and the second benchmark. The other four do not reproduce it.

The two evaluation choices behind that failure are of different kinds. Which matched point is reported matters because realised capability is still moving: Section 6.5 finds no architecture term distinguishable from zero once loss is adjusted for continuously, while the loss term carries an effect of the same order as the architecture contrast itself. That is consistent with capability accounting for much of the single-point contrast, but it is weaker than a causal claim, since with 21 snapshots from four runs and one seed a non-significant coefficient may equally reflect limited snapshot-level information. Which scoring rule is applied matters because the rules disagree on the token set and, decisively, on which member of a pair counts as stereotypical. Section 6.6 attributes the sign reversal to that convention alone, and neither convention is indefensible, which is the central difficulty: a comparison surviving under only one of them is not yet a finding.

Retrieval and classification pipelines routinely select one pre-trained encoder over another, sometimes partly on fairness grounds. The evidence here is that such a comparison can reverse under three choices that are seldom reported: the matched capability level, the scoring rule, and the direction convention. A comparison that does not report all three is not yet decisive.

8 Limitations

Training uses a single seed per architecture, so the four runs are not replicated. Every interval reported here bootstraps over benchmark items and so quantifies item sampling alone, not variation between training runs. The ordering of the encoders may not be stable under re-training with another seed. This is the study's largest limitation, and it applies to the 21 snapshots collectively, since they are successive checkpoints of only four runs.

The analyses across matched points, the continuous loss adjustment, the scoring-rule decomposition, the per-category comparisons and the equivalence check were performed after the single-point result had been examined, and are reported as exploratory. Only the three contrasts at the deepest matched point were specified in advance.

The encoders reach roughly 2.18 nats at the deepest matched point, well short of convergence at this size, and the effect is still changing there. The Spearman correlation between matched validation loss and the mean *CrowS-Pairs* effect, taken across each encoder's own distinct matched points, is −0.80, −0.90 and −0.71 for *VanillaBERT*, *ALBERTLoopedBERT* and *HyperloopBERT* over 4, 5 and 7 points. *LoopedBERT* does not follow that trend, at +0.30 over 5 points. Whether these contrasts differ for fully trained encoders is untested here.

Two benchmarks are used, both of the correlation type in the sense of Wang et al. [20], and both scored from the same masked language modelling head, so agreement between them is weaker evidence than agreement between independent measurement approaches would be. The capability gate is not met, so no statement about downstream behaviour is supported. Both instruments carry documented item-quality problems [2], of which the eight unalignable pairs in Section 6.6 are an instance. Finally, the scoring-rule decomposition isolates the direction convention at one matched point, and does not establish that the same attribution holds elsewhere; neither the other matched points nor *StereoSet* was rescored under the official rule.

The evaluation is English-only. An India-centric instrument covering caste and religion [9] was planned, but the mirror available at the time could not be used with an English-only tokeniser. Given the regional focus of this venue that omission is stated rather than passed over, and extending the protocol to it is the first direction this work would pursue.

9 Reproducibility

Data are as in Section 3, with a checksum and block-count manifest verified at the start of every run. Architectures are as in Table 2, trained with AdamW, peak learning rate 3×10^{-4} (HyperloopBERT 1.5×10^{-4}), batch size 512, 10% warmup, gradient clipping 1.0, bfloat16 autocast, one seed (42), on one H100 GPU. Attention uses the memory-efficient exact implementation of Dao et al. [3], which the run log confirms was active throughout. Snapshots are selected by first crossing of a pre-set validation loss target, yielding the 21 distinct snapshots analysed here. Scoring is 32-bit under all three rules; the *StereoSet* replication and the official-rule rescoring both rerun on CPU from the same snapshots, and the former reproduces the stored *CrowS-Pairs* item scores to within 10^{-6}. Inference is sign-flip permutation with $m = 10^4$ draws and Phipson–Smyth $(b + 1)/(m + 1)$ p-values [15], 10^4-resample percentile item bootstraps, Holm correction within each three-contrast family, and cluster-robust OLS under the four dependence models of Section 6.5. Analyses run under Python 3.10 with NumPy 2.2, pandas 2.3, SciPy 1.15 and statsmodels 0.14, with all random seeds fixed; re-running the released scripts, which include the scoring-rule decomposition, reproduces every reported number exactly. Code, per-item scores for every snapshot, the masked-position predictions behind Table 5, and the analysis scripts are available at https://anonymous.4open.science/r/HyperloopBert/.

10 Conclusion

Four encoders differing in how far they reuse weights across depth were pre-trained from scratch on identical data and compared at matched validation loss over 21 distinct snapshots. At the deepest matched point on *CrowS-Pairs*, two of the three weight-reusing encoders record a lower stereotype effect than the unshared baseline, both surviving Holm correction at $p_{\mathrm{Holm}} = 0.0003$. That result does not survive the checks that follow. It holds at one of the five matched points available for the looped encoder, becomes indistinguishable from zero once realised loss is adjusted for continuously under four dependence models, and survives correction under neither alternative scoring rule. One rule reverses its sign, and that reversal is traced to a convention governing 218 of the 1508 pairs rather than to the token set or the aggregation. On *StereoSet* the direction replicates but only one of three contrasts retains significance, for a different encoder. *CrowS-Pairs* aggregates conceal sign reversals across four of its nine categories, and all four encoders perform at chance on *WinoBias*.

The conclusion is methodological rather than architectural. A bias comparison resting on one matched checkpoint and one scoring definition can reverse under either choice, and under conventions inherited unexamined from a reference implementation. Such comparisons should report sensitivity across matched capability levels and scoring rules, and state the direction convention, before attributing any difference to architecture.

Generative AI Use Disclosure

ChatGPT was used to correct grammatical mistakes and to improve sentence structure and readability. The authors reviewed the content and take full responsibility for it.

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